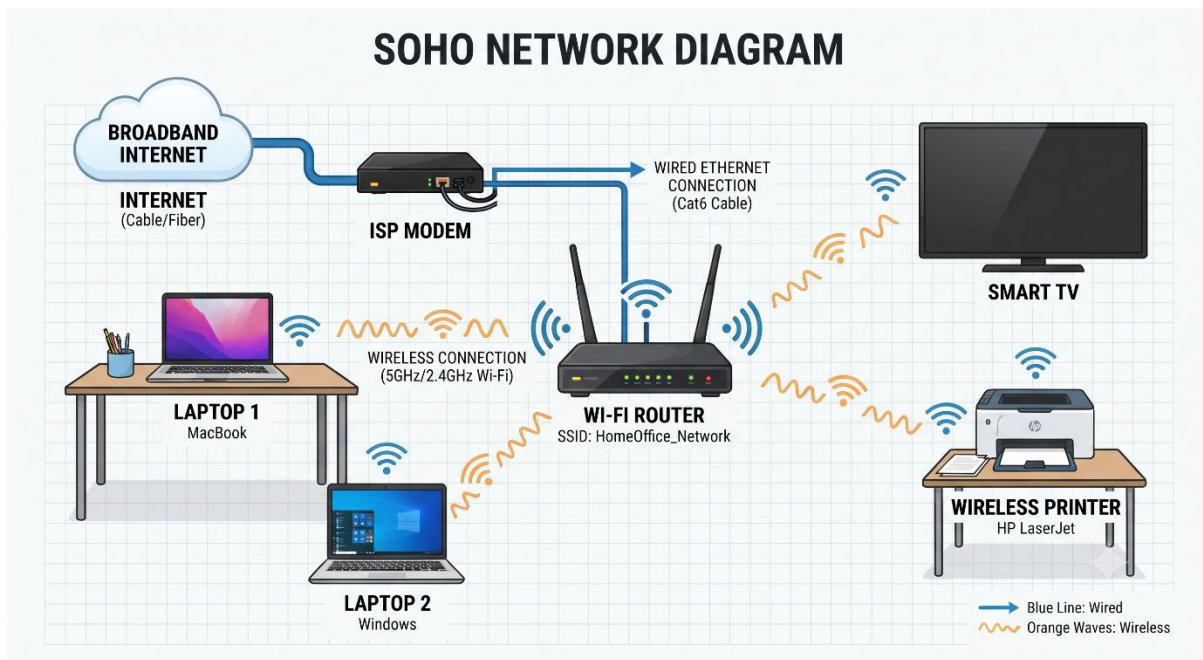


ASSIGNMENT 1

Bulid a SOHO network

Task (1):



Task (2):

To build the exact network shown in the diagram for your Small Office/Home Office (SOHO), here are the recommended hardware components and model suggestions easily available on Indian e-commerce platforms like Amazon and Flipkart:

1. Core Network Hardware:

- **Broadband Internet Connection**
 - Suggestion: Airtel Xstream Fiber or JioFiber broadband connection (typically supplied with an Optical Network Terminal / ISP Modem).

- **Wi-Fi Router**

- Suggestion: TP-Link Archer AX12 AX1500 Wi-Fi 6 Router or TP-Link Archer C6 AC1200. These are budget-friendly, highly reliable dual-band routers widely used in Indian homes and small offices for seamless wireless coverage.

2. End-User Devices:

- **Two Laptops:**

- Apple MacBook Air (M1/M2) for Laptop 1 and a Windows-based laptop like an HP Laptop 15 or Lenovo IdeaPad for Laptop 2. Both connect wirelessly via 5GHz/2.4GHz Wi-Fi.

- **Smart TV:**

- Xiaomi X Series 43-inch 4K Smart TV or Samsung Crystal 4K UHD TV, which connect directly to your network over Wi-Fi.

3. Cables and Accessories:

- **Wired Ethernet Connections:**

- UGREEN High-Speed Cat 6 Ethernet Cable or D-Link Cat6 UTP Patch Cord. You will need at least two patch cables: one to connect the ISP Modem to the Wi-Fi router, and an optional backup cable if you prefer a wired link for your printer or computer.

Task (3):

Configuring your Wi-Fi router correctly is essential for securing your SOHO network against unauthorized access. Follow these steps to set up your router's core security features.

Steps:

1. Connect to the Router Panel:

- Connect your computer to the router via an Ethernet cable or its default Wi-Fi (found on the sticker underneath).
- Open a web browser, type the router's IP address (such as 192.168.0.1 or 192.168.1.1), and log in using the default credentials printed on the device.

2. Change the Admin Password:

- Go to the Administration, System Tools, or Password settings menu. Enter a strong, unique new password to secure your router's settings page, then save.

3. Set a Unique Network Name (SSID):

- Navigate to the Wireless or Wi-Fi Settings section. Type your preferred name for your home office network into the SSID field and save the changes.

4. Enable WPA2 Security:

- Under the wireless security settings, select WPA2-PSK (or WPA2/WPA3 mixed mode) as your encryption type. Enter a secure Wi-Fi password and save your settings.

Task (4):

Device	MAC Address (Example)	IP Address Type	Assigned IP Address
Wi-Fi Router (Gateway)	10:A4:BE:01:23:45	Static	192.168.1.1
Smart TV	AA:BB:CC:11:22:33	Static	192.168.1.50
Wireless Printer	D4:3D:7E:AA:BB:CC	Static	192.168.1.51
Laptop 1 (MacBook)	70:F1:A1:22:33:44	DHCP (Dynamic)	192.168.1.101
Laptop 2 (Windows)	48:E2:44:55:66:77	DHCP (Dynamic)	192.168.1.102

- **Static IP:** A permanent address manually set for devices like the TV and printer so they never change, making them easy to find.
- **DHCP (Dynamic IP):** A temporary address automatically handed out by the router to portable devices like laptops whenever they connect.

Task (5):

Three common causes when a single laptop loses internet connectivity in a SOHO network, along with steps to diagnose and fix them.

Cause 1: Incorrect Wi-Fi Password or Authentication Error

If the network password was recently changed or entered incorrectly, the laptop will fail to authenticate with the router.

Solution:

- Go to your laptop's Wi-Fi settings, select "Forget" or "Delete" the home office network, search for the network again, and reconnect by entering the correct WPA2 password.

Cause 2: IP Address Assignment (DHCP) Failure:

If the laptop fails to obtain a valid IP address from the router's DHCP server, it will remain connected to the Wi-Fi without internet access (often shown as a "No Internet" or "Limited Access" warning).

Solution:

- **On Windows:** Open Command Prompt and type **ipconfig /release** followed by **ipconfig /renew**.
- **On Mac:** Open System Settings > Network > Wi-Fi, click Details, and select **Renew DHCP Lease**.

Cause 3: Disabled or Glitched Wi-Fi Adapter:

The laptop's internal wireless card or network adapter may have crashed or been accidentally disabled.

Solution:

- Toggle **Airplane Mode** on for 10 seconds and turn it back off to reset all wireless radios. If that fails, restart the laptop completely.

